Emanuel Azcona, Ph.D.

San Francisco, CA   emanazcona@gmail.com   linkedin.com/in/emanuelazcona/

SUMMARY

Data scientist and technical lead with experience in multimodal AI, generative models, graph-based machine learning, computer vision, and computational neuroscience. Experienced in developing production-scale multimodal AI systems that transform images, video, and product data into actionable business intelligence through foundation models, representation learning, and novel measurement frameworks. Passionate about using data and AI to understand human perception, consumer behavior, and decision-making at scale.

INDUSTRY EXPERIENCE

Lead Data Scientist | *Nike, Inc. – Product Creation & Merchandising Data Science*

(Feb. 2025 – Present) · San Francisco, CA (Remote)

- Fine-tuned and adapted **TRELLIS2, a state-of-the-art (SOTA) generative 2D-to-3D foundation model**, using curated proprietary product datasets to improve geometry reconstruction, material fidelity, and generalization across footwear product categories.
- Enabled **transfer learning strategies that preserved pretrained 3D priors** through selective component tuning and layer-freezing approaches, enabling efficient domain adaptation while maintaining generalizable generative capabilities.
- Helped **advance experimental 3D generative AI research into reusable product creation workflows**, supporting broader exploration of AI-native design, digital product representation, and synthetic asset generation.
- Applied **multimodal foundation models (TRIBEv2) to generate synthetic neural biomarkers** for image and video marketing assets, introducing new quantitative signals for evaluating content through the lens of human perception and cognitive response.
- Developed scalable pipelines for estimating **fMRI-based activation profiles**, enabling large-scale analysis of marketing impressions, visual saliency, engagement potential, and attention-driving content characteristics.
- Led the development of multimodal AI pipelines that **transformed visual product assets into quantitative business intelligence**, generating perception-based color measurements across **90K+ footwear product measurements used across 20+ global teams.**
- Designed the **human-in-the-loop (HITL) evaluation strategy**, combining human-labeled ground truth data, VLM-as-a-judge validation, inter-rater reliability, and perception-aligned quantitative metrics to establish production-quality standards.
- Built large-scale multimodal inference and evaluation pipelines spanning product intelligence, consumer perception modeling, and physical activity recommendation systems using Databricks, PySpark, Python, and MLflow.

Senior Data Scientist | *Nike, Inc. – Product Creation & Merchandising Data Science*

(Feb. 2023 – Feb. 2025) · Chicago, IL (Remote)

- Led development of a **product similarity platform powered by HNSW vector search**, enabling scalable substitute recommendations for assortment planning and merchandising workflows.
- Designed **embedding-based product representations** that captured visual and semantic relationships between products, improving product discovery and similarity retrieval beyond traditional attribute-based approaches.
- Enhanced recommendation systems through **fusion of behavioral engagement signals and perceptual color features**, improving personalization relevance beyond discrete product color classifications.
- Partnered with business and technical stakeholders to **translate machine learning research into production solutions** supporting product creation and marketplace analytics.

Data Scientist | *Nike, Inc. – Product Creation & Merchandising Data Science*

(Sept. 2022 – Jan. 2023) · Seattle, WA (Remote)

- Developed **color-theory-grounded computer vision pipelines** to quantify product color composition from imagery, establishing the foundation for future multimodal product intelligence initiatives.
- Co-developed business logic with merchandising and regional analytics teams to label individual footwear products by perceived color blocking to enable colorway preference segmentation.

Data Scientist & Graduate Date Science Intern | *Nike, Inc. – Sport Activity Data Science*

(June 2021 – Sept. 2022) · Evanston, IL (Remote)

- Developed **graph neural network (GNN)-based recommendation systems** for the Nike Training Club platform, leveraging biometric, behavioral, and engagement data to improve workout personalization.
- Applied **representation learning techniques** to generate user and workout embedding visualizations and capture latent engagement patterns and workout affinities from historical user-workout interactions.
- Identified actionable insights from **embedding-space analysis**, informing workout sequencing, content strategy, and retention-focused product improvements.

Artificial Intelligence Intern | ***Stats Perform***
(June – Aug. 2018) · Chicago, IL

- Built scalable pipelines for processing **large-scale spatiotemporal player-tracking data** from professional football competitions using MongoDB and AWS tools.
- Designed **temporal graph convolutional autoencoders** to learn compact latent representations of multi-agent movement dynamics, achieving sub-millimeter reconstruction accuracy.
- Applied **geometric deep learning techniques** to model complex player interactions, enabling downstream research in prediction, simulation, and multi-agent behavior analysis.

ACADEMIC EXPERIENCE

Adjunct Associate Faculty | ***Columbia University***
(Sept. 2025 – Present) · San Francisco, CA (Remote)

- Provided strategic and technical mentorship for industry-sponsored capstone projects, guiding students through the end-to-end development of machine learning and analytics solutions for real-world business problems.
- Advised student teams on customer segmentation, conversion modeling, experimentation, churn prediction, and the translation of analytical findings into actionable business recommendations.
- Mentored students on machine learning methodology, data engineering, statistical rigor, and executive communication, helping bridge technical execution with business impact.
- Partnered with faculty to deliver course instruction and ensure successful execution of cross-functional, industry-driven projects.

Ph.D. Candidate & Research Assistant | ***Northwestern University, Image & Video Processing Lab***
(Sept. 2017 – June 2022) · Evanston, IL

- Conducted interdisciplinary research spanning **computational neuroscience, representation learning, geometric deep learning, and medical imaging**, developing machine learning methods to model complex cognitive and neurological phenomena.
- Authored and presented multiple peer-reviewed publications applying **graph neural networks (GNNs), multimodal learning, and neuroimaging analytics** to Alzheimer's disease characterization and human behavior modeling.
- Developed **GNN and variational autoencoder (VAE) architectures** to learn interpretable representations of brain structure, cognitive processes, and neurological disease progression.
- Built a **GNN-based diagnostic classifier achieving 96.35% accuracy** in distinguishing Alzheimer's patients from healthy controls across 300+ subjects while preserving interpretability through graph-level and regional analyses.

EDUCATION

- **Ph.D. Electrical Engineering, 2022,** *Northwestern University*
  Dissertation: *Geometric Deep Learning in Neuroimaging and Human Reward Behavior*
- **M.S. Electrical Engineering, 2019,** *Northwestern University*
- **B.S. Electrical Engineering, 2017, *cum laude,*** *New York University*
  Senior Thesis: *Predicting NBA Playoff Contention Through Individual Player Performance*

TECHNICAL SKILLS

Languages:	Python, SQL, C++, Matlab, Bash, LaTeX, HTML/CSS
DS & ML:	PyTorch, Scikit-learn, Pandas, Tensorflow/Keras, PyTorch Geometric (PyG), Matplotlib/Seaborn, Docker
Data & Cloud:	PySpark, AWS (S3, Sagemaker Studio), Databricks, MongoDB, Google Colab
Domains:	Machine learning, medical imaging , graph neural networks, recommender systems, embedding analysis, computer vision, image processing, convolutional neural networks, transformers, vision language models (VLMs), prompt engineering

COMMUNICATION & LEADERSHIP

Languages:	Full professional proficiency in English and Spanish (bilingual).
Documentation Excellence:	Commended by peers for delivering high-impact, cohesive technical documentation that bridges the gap between machine learning (ML) research and scalable industry application.
Workshop/Demo Facilitation:	Architect and deliver technical workshops on multimodal AI (VLM) and GenAI, prioritizing clarity for both technical and non-technical audiences.